

Day 3 - Mockup Exams AI-900

1. Which artificial intelligence (AI) technique should be used to extract the name of a store from a photograph displaying the store front?

- a. Image classification
- b. Natural Language Processing (NLP)
- c. Optical Character Recognition (OCR)

OCR provides the ability to detect and read text in images. NLP is an area of AI that deals with identifying the meaning of a written or spoken language, but not detecting or reading text in images. Image classification classifies images based on their contents.

2. Which computer vision solution provides the ability to identify a person's age based on a photograph?

- a. Facial detection
- b. Image classification
- c. Object detection

Facial detection provides the ability to detect and analyze human faces in an image, including identifying a person's age based on a photograph. Image classification classifies images based on their contents. Object detection provides the ability to generate bounding boxes identifying the locations of different types of vehicles in an image.

3. Which Azure Cognitive Service for Language feature allows you to analyze written articles to extract information and concepts, such as people and locations, for classification purposes?

- a. Content Moderation
- b. Key Phrase Extraction
- c. Named Entity Recognition

Named entity recognition can identify and categorize entities in unstructured text, such as people, places, organizations, and quantities, and is suitable to support the development of an article recommendation system. Key phrase extraction, Content Moderator.

4. Which type of machine learning algorithm assigns items to a set of predefined categories?

- a. Classification
- b. Clustering
- c. Regression
- d. Supervised

Classification algorithms are used to predict a predefined category to which an input value belongs. Regression algorithms are used to predict numeric values. Clustering algorithms group data points that have similar characteristics. Unsupervised learning is a category of learning algorithms that includes clustering, but not regression or classification.

5. Which feature of computer vision involves associating an image with metadata

that summarizes the attributes of the image?

- a. Content organization
- b. Detecting image types
- c. Tagging

Tagging involves associating an image with metadata that summarizes the attributes of the image. Detecting image types involves identifying clip art images or line drawings. Content organization involves identifying people or objects in photos and organizing them based on the identification.

6. You need to use Azure Machine Learning to train a regression model. What should you create in Machine Learning studio?

- a. A job
- b. A workplace
- c. An Azure container instance
- d. An Azure Kubernetes Service (AKS) cluster

7. Which artificial intelligence (AI) technique serves as the foundation for modern image classification solutions?

- a. Anomaly detection
- b. Deep learning
- c. Linear regression
- d. Multiple linear regression

Modern image classification solutions are based on deep learning techniques. Anomaly detection analyzes data collected over time to identify errors or unusual changes. Both linear regression and multiple linear regression use training and validating predictions to predict numeric values, so they are not part of image classification solutions.

8. Which tool provides the easiest way to create a knowledge base for Azure Bot Service?

- a. The azure machine learning studio
- b. The azure portal
- c. The language studio

The Language Studio provides the easiest way to create a knowledge base for Azure Bot Service. While it is possible to create a knowledge base by using the features of the Machine Learning studio, this is not the simplest method to accomplish the given objective. The Azure portal allow for creating the Language resource, but will not create a knowledge base for Azure Bot Service.

9. Which type of artificial intelligence (AI) workload provides the ability to classify individual pixels in an image depending on the object that they represent?

- a. Image classification
- b. Object detection
- c. Semantic segmentation

Semantic segmentation provides the ability to classify individual pixels in an image depending on the object that they represent. The other answer choices also process images, but their outcomes are different.

10. Which artificial intelligence (AI) workload scenario is an example of natural language processing (NLP)?

- a. extracting key phrases from a business insights report
- b. identifying objects in landscape images
- c. predicting whether customers are likely to buy a product based on previous purchases

Extracting key phrases from text to identify the main terms is an NLP workload. Predicting whether customers are likely to buy a product based on previous purchases requires the development of a machine learning model. Identifying objects in landscape images is a computer vision workload.

11. You are exploring solutions to improve the document search and indexing service for employees. You need an artificial intelligence (AI) search solution that will include searching text in various types of documents, such as images. Which type of AI workload is this?

- a. Anomaly detection
- b. Computer vision
- c. Conversational AI
- d. Data mining

Data mining workloads primarily focus on the searching and indexing of data. The computer vision service can be used to extract information from images, but it is not a search and indexing solution. Conversational AI is part of natural language processing (NLP) and facilitates the creation of chatbots. Anomaly detection is not used for searching and indexing data. This is used to detect outliers in data, such as unusual credit card activity for the detection of fraud.

12. Which principle of responsible artificial intelligence (AI) has the objective of ensuring that AI solutions benefit all parts of society regardless of gender or ethnicity?

- a. Accountability
- b. Inclusiveness
- c. Privacy and Security

The inclusiveness principle is meant to ensure that AI solutions empower and engage everyone, regardless of criteria such as physical ability, gender, sexual orientation, or ethnicity. Privacy and security and accountability do not discriminate based on these criteria, but also do not emphasize the significance of bringing benefits to all parts of the society.

13. Which Azure Cognitive Service for Language feature can be used to analyze online user reviews to identify whether users view a product positively or negatively?

- a. Key Phrase Extraction
- b. Language Detection
- c. Named Entity Recognition
- d. Sentiment Analysis

Sentiment analysis provides sentiment labels, such as negative, neutral, and positive, based on a confidence score from text analysis. This makes it suitable for understanding user sentiment for product reviews. The named entity recognition, key phrase extraction, and language detection features cannot perform sentiment analysis for product reviews.

14. Which part of speech synthesis in natural language processing (NLP) involves breaking text into individual words such that each word can be assigned phonetic sounds?

- a. Key Phrase Extraction
- b. Tokenization
- c. Transcribing

Tokenization is part of speech synthesis that involves breaking text into individual words such that each word can be assigned phonetic sounds. Transcribing is part of speech recognition, which involves converting speech into a text representation. Key phrase extraction is part of language processing, not speech synthesis.

15. Which feature of the Language service includes functionality that returns links to external websites to disambiguate terms identified in a text?

- a. Entity recognition
- b. Key phrase extraction
- c. Language detection
- d. Sentiment Analysis

Entity recognition includes the entity linking functionality that returns links to external websites to disambiguate terms (entities) identified in a text. Key phrase extraction evaluates the text of a document and identifies its main talking points. Language detection identifies the language in which text is written. Sentiment analysis evaluates text and returns sentiment scores and labels for each sentence.

16. You have a dataset that contains information about taxi journeys that occurred during a given period. You need to train a model to predict the fare of a taxi journey.

What should you use as a feature?

- a. the number of taxi journeys in the dataset
- b. the trip distance of individual taxi journeys
- c. the fare of individual taxi journeys
- d. the trip ID of individual taxi journeys

The label is the column you want to predict. The identified Features are the inputs you give the model to predict the Label. Example: The provided data set contains the following columns: vendor_id: The ID of the taxi vendor is a feature. rate_code: The rate type of the taxi trip is a feature. passenger_count: The number of passengers on the trip is a feature. trip_time_in_secs: The amount of time the trip took. You want to predict the fare of the trip before the trip is completed. At that moment, you don't know how long the trip would take. Thus, the trip time is not a feature and you'll exclude this column from the model. trip_distance: The distance of the trip is a feature. payment_type: The payment method (cash or credit card) is a feature. fare_amount: The total taxi fare paid is the label.

Reference: <https://docs.microsoft.com/en-us/dotnet/machine-learning/tutorials/predict-prices>

17. You need to predict the sea level in meters for the next 10 years. Which type of machine learning should you use?

- a. Classification
- b. Regression
- c. clustering

In the most basic sense, regression refers to prediction of a numeric target. Linear regression attempts to establish a linear relationship between one or more independent variables and a numeric outcome, or dependent variable. You use this module to define a linear regression method, and then train a model using a labeled dataset. The trained model can then be used to make predictions.

Reference:

<https://docs.microsoft.com/en-us/azure/machine-learning/studio-module-reference/linear-regression>

18. Which service should you use to extract text, key/value pairs, and table data automatically from scanned documents?

- a. Form Recognizer

- b. Text Analytics
- c. Language Understanding
- d. Custom Vision

Accelerate your business processes by automating information extraction. Form Recognizer applies advanced machine learning to accurately extract text, key/value pairs, and tables from documents. With just a few samples, Form Recognizer tailors its understanding to your documents, both on-premises and in the cloud. Turn forms into usable data at a fraction of the time and cost, so you can focus more time acting on the information rather than compiling it.

Reference: <https://azure.microsoft.com/en-us/services/cognitive-services/form-recognizer/>

19. A medical research project uses a large anonymized dataset of brain scan images that are categorized into predefined brain haemorrhage types. You need to use machine learning to support early detection of the different brain haemorrhage types in the images before the images are reviewed by a person. This is an example of which type of machine learning?

- a. Clustering
- b. Regression
- c. classification

This is classification. There are multiple reasons it's classification

- The training data is already tagged as with the correct type of hemorrhage

- Classification can be done for more than two classes (which people seem to not realize based on the comments)

- You do clustering on a group of inputs. For example, the scans of 10 people. You can't cluster a single input. Clearly you get a new scan of a new patient and you want to know what that scan shows, you don't have a group of scans to cluster.

- Clustering gives NO labels. You just get groups and don't know what the label is, but in this question it's very clear they want to know the label that belongs to the new scan -> classification

Reference:

<https://docs.microsoft.com/en-us/learn/modules/create-classification-model-azure-machine-learning-designer/introduction>

20. When training a model, why should you randomly split the rows into separate subsets?

- a. to train the model twice to attain better accuracy
- b. to train multiple models simultaneously to attain better performance
- c. to test the model by using data that was not used to train the model

First for training then evaluation. He is referring to the famous train_test_split that everyone uses to split the dataset into train and test sets. You lose statistical power by estimating on a subset n like 50 percent of original N . That's the price you pay for splitting data. For example, a normally independently distributed (n.i.d) x has an estimator of arithmetic mean $\bar{X}$ whose variance is inversely related to N . Bigger sample is better for accuracy of a simple arithmetic mean.

